21 Malik is going to throw a fair coin 50 times.

(a) Write down an estimate for the number of times the coin will land on heads.

Fair coin: 50 × 1/2 = 25 heads expected.

1 mark
25

.....

(1)

Paula and Simon are trying to find out if a different coin is biased.

Paula throws this coin 10 times.
She records the number of times the coin lands on heads.

Simon throws the same coin 100 times.
He records the number of times the coin lands on heads.

(b) Whose results will be more useful in deciding if the coin is biased?
Give a reason for your answer.

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(1)

(Total for Question 21 is 2 marks)

Simon's 100 throws are more useful.

A larger sample gives a more reliable estimate of bias.

1 mark
Simon
100 trials is larger

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